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Is this real life? – Exploring leadership communication on Twitter/X

Christina Strauss[a], Michael Dominic Harr[a] and Torsten Pieper[b]

[a]Chair of Business Informatics and Integrated Information Systems, University of Duisburg-Essen, Essen, Germany; [b]Belk College of Business, University of North Carolina at Charlotte, Charlotte, NC, USA

ABSTRACT

As central actors in corporate communications, CEOs embody the values and goals of their organisation. With ongoing digitalisation and the proliferation of social media in both private and business environments, leadership communication has undergone profound change. In this vein, research increasingly acknowledges the role of 'social CEOs' who engage with their stakeholders in genuine communications over social media. However, research on leaders' distinct social media communication remains scarce. To address this gap, our study explores leadership communication via social media in comparison to traditional (non-social media) communication. Based on tweets and non-social media documents from 88 CEOs of worldwide leading companies, our results indicate that leaders use a distinct social media framing that is significantly different from their non-social media framing. Our findings have theoretical and practical implications for the understanding of leaders' framing behaviour as well as the resulting social media communication.

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1. Introduction

As the main representatives of their firm, Chief Executive Officers (CEOs) are vital for companies' internal and external communication (Chen, Diao, and Zhang 2011; Tsai and Men 2017). This relevance has been further intensified by digitalisation. The spread of digital technologies in general and social media (SM) in particular (Chen, Rui, and Whinston 2021) has led to a transformation of leadership and leadership communication (Cortellazzo, Bruni, and Zampieri 2019). In this context, SM offers corporate leaders (here defined as CEOs) a wide variety of opportunities not only to share information and access resources (Castelló, Etter, and Årup Nielsen 2016; Heavey et al. 2020), but also to exert social influence (see also Banks et al. 2022; Blankespoor 2018). Thereby, they act as 'social CEOs' (Tsai and Men 2017, 58) who actively engage and communicate with their stakeholders.

Consequently, SM plays a crucial role when discussing digital leadership (Avolio, Kahai, and Dodge 2001), being defined as 'a social influence process embedded in both proximal and distal contexts mediated by [advanced information technology] that can produce a change in attitudes, feelings, thinking, behavior, and performance' (Avolio et al. 2014, 107). In this vein, according to framing theory (Entman 1993), CEOs use SM not only to disseminate information but also to engage in special tactics, i.e. presenting issues as well as themselves as a person in a specific way (Alghawi, Yan, and Wei 2014; Heavey et al. 2020; Kim et al. 2015; Matthews et al. 2022; Men and Tsai 2016; Tsai and Men 2017). Consequently, SM framing enables leaders to influence perceptions, opinions, attitudes, and even actions of their followers (Ante 2023; Kim et al. 2015; Matthews et al. 2022; Tsai and Men 2017).

Despite its increasing relevance in practice, research on CEOs' SM communication and especially their framing remains scarce (Matthews et al. 2022) or relies on experimental studies over a short time horizon (e.g. Alghawi, Yan, and Wei 2014; Chen, Rui, and Whinston 2021; Men and Tsai 2016; Tsai and Men 2017). However, due to the relevance of social and emotional aspects of SM as well as its dependence on societal developments that can hardly be simulated, investigating the phenomenon under in vivo conditions is vital (Strauss, Harr, and Pieper 2024).

From a methodological perspective, SM holds great promise for addressing this requirement since, nowadays, an increasing amount of real-life data from

CONTACT Christina Strauss christina.strauss@ris.uni-due.de Chair of Business Informatics and Integrated Information Systems, University of Duisburg-Essen, Universitätsstraße 9, 45141, Essen, Germany

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corporate leaders' SM use is available (Liu, Chambers, and Moore 2023) – for example, in the form of corporate leaders' tweets, LinkedIn posts, blog entries, and other publicly accessible SM communications. These data allow to unravel context-specific intra- (i.e. within leaders) and interindividual (i.e. between leaders and/or with stakeholders) developments essential for understanding leaders' communication on SM (Liu, Chambers, and Moore 2023).

This understanding is important for several reasons. First, from a methodological point of view, the medium of SM presents unique characteristics that warrant careful examination. Accounting for this uniqueness is particularly important due to SM's increasing prevalence in professional environments (Matthews et al. 2022), where individuals are under increased scrutiny (e.g. Ante 2023; Browne 2021). Second, focusing on leaders is particularly important because their communication has wide-ranging impacts on interpersonal relationships (Hill, Kang, and Seo 2014), organisational dynamics (Schein 2010), and economic outcomes (Wong, Ormiston, and Haselhuhn 2011). For instance, leaders' communication affects corporate culture and strategy, contributes to potential competitive advantages, shapes the public image of both the leader and the company, and influences the broader social landscape (Heavey et al. 2020; Love, Lim, and Bednar 2017; Niburski and Niburski 2020).

In this paper, we shed light on CEOs' communication via SM by collecting and analysing data on leaders' framing – meaning the process of highlighting or emphasising particular aspects within communication to shape a certain understanding (Entman 1993) – on the SM platform Twitter.[1] Twitter is a SM platform for microblogging which allows users to broadcast short messages to a (potentially) broad, subscribing audience (Huang and Yeo 2018). In further detail, we focus on the following research questions (RQs):

RQ1: 'How do leaders frame communication via Twitter?'

RQ2: 'How does this framing differ from non-SM contexts?'

Based on a unique dataset consisting of CEOs' Twitter data, complemented by personal information (e.g. age, gender) and transcripts of leaders' interviews, public speeches, letters from the CEO, publications, and messages to the employees, we analyse leadership communication of 88 corporate CEOs. In this vein, we omit the 'ritualized use of questionnaires' (Fischer et al. 2020, 1) in leadership research and employ novel, qualitative, and technology-based research templates (McDermott 2023).

We investigate different, scientifically validated linguistic framing dimensions (Pennebaker Conglomerates 2022; Tausczik and Pennebaker 2010) in leaders' communications across both SM and non-SM contexts, to determine whether leaders' framing differs by communication channel and if it changes over time. Additionally, we conduct a topic model analysis to uncover CEOs' most frequently used words and topics in both SM and non-SM contexts.

For this purpose, we focus on the SM platform Twitter due to its high standardisation, controllability, and availability (Bamberger and Pratt 2010; Heavey et al. 2020; Tur, Harstad, and Antonakis 2022). In addition, as prior studies have also been conducted with Twitter data, using the same data source enables us to connect and compare our results to previous findings (e.g. Bilinski 2022; Chen, Rui, and Whinston 2021; Lee, Hwang, and Chen 2017; Niburski and Niburski 2020).

Our work makes several theoretical and practical contributions to leadership and corporate communications. First, we demonstrate the contextuality of leadership communication by identifying distinct SM framing. In this vein, we adopt a contingency perspective to examine the context-specific (i.e. SM vs. non-SM) framing employed by CEOs across various communication mediums, as delineated by Cortellazzo, Bruni, and Zampieri (2019). Thereby, we address the recognised 'context deficit' (Johns 2024, 1) gap in existing leadership research. Second, we show that leaders' communication on SM depends on the medium rather than the person (i.e. 'to what extent [leaders' framing] is different in a digital era' (Banks et al. 2022, 1)). Third, following a call from Matthews et al. (2022), who argued for future research on leaders' intertemporal framing behaviour on SM, we find that framing tactics used by leaders change over time. This suggests that leadership on SM involves continuously adapting framing strategies in response to topics targeted, organisational goals, or audience expectations.

Furthermore, our findings emphasise the importance of context-specific and intertemporal analyses of leaders' communication. Hence, we address the critical need to understand how leaders adapt their communications over time and thereby advance the ongoing discourse on leadership framing in corporate communications, which has been heavily dominated by political contexts so far (Matthews et al. 2022).

The remainder of this paper is structured as follows. The following section presents the theoretical underpinnings of our study, focusing on SM and framing as a leadership research context. The method section describes our sample as well as the data collection and analysis

procedures. The results section highlights findings from our study. The final discussion section summarises the main insights, addresses the limitations of our work, and offers implications and avenues for future research.

2. Theoretical background

2.1. Social media as a channel for leadership communication

Social media, being defined as 'a group of Internet-based applications that build on the ideological and technological foundations of Web 2.0, and that allow the creation and exchange of User Generated Content' (Kaplan and Haenlein 2010, 61), has grown significantly in recent years (Chen, Rui, and Whinston 2021; Matthews et al. 2022); not only in user numbers, but also in terms of relevance in a wide variety of areas (e.g. Browne 2021; Chen, Rui, and Whinston 2021; Niburski and Niburski 2020). It is used to socialise, interact with firms or brands, look for jobs, network or even do business (Aichner et al. 2021), thereby covering nearly every aspect of life.

Nowadays, CEOs use SM in multiple ways as well; for example to provide sensemaking (Pye 2005; Schmidt and Van Dellen 2022), access public intelligence (Constantinides 2014), gain resources or support (Castelló, Etter, and Årup Nielsen 2016; Schmidt and Van Dellen 2022), collaborate and co-create (Leonardi, Huysman, and Steinfield 2013; Seidl and Werle 2018), or increase stakeholder influence (Cabano, Li, and Jiménez 2024; Heavey et al. 2020; Yadav et al. 2024). As representatives of their company, leaders' public image is intertwined with that of the company (Love, Lim, and Bednar 2017), which especially manifests in SM (Etter, Ravasi, and Colleoni 2017) as an instrument to 'shape their public image and that of their company' (Malhotra and Malhotra 2016, 74).

Research has shown that leaders adopt diverse approaches to engagement on SM. Schmidt et al. (2025), for example, highlighted that the content of shared posts can vary. Their findings indicated that entrepreneurial CEOs often utilise SM platforms like Twitter to share information and links, or product advertisements. The same research also revealed gender differences between male and female leaders, noting that female leaders tended to share personal opinions more frequently than their male counterparts. This varying use of SM is further supported by other research identifying distinct user-types of CEOs on Twitter based on their posting behaviour (Malhotra and Malhotra 2016).

Regardless of the specific approach taken, leaders can actively leverage SM to influence the perceptions, opinions, attitudes, and even actions of their followers (Alghawi, Yan, and Wei 2014; Ante 2023; Browne 2021; Kim et al. 2015; Niburski and Niburski 2020) by 'sharing interpretations of events and direct attention to issues and causes from a distance' (Heavey et al. 2020, 1495). For example, it was shown that Twitter posts by Donald Trump in the context of the COVID-19 pandemic led to an increase in Google search queries for unproven treatment methods as well as their sales on Amazon (Niburski and Niburski 2020) or that posts by Elon Musk significantly influenced the trading value of cryptocurrencies (Ante 2023).

It is important to note that this type of framing does not only happen in the context of public events. Donald Trump, for example, also used strategic framing by emphasising that 'the number of followers on Twitter was an important indicator of support' (Schmidt and Van Dellen 2022, 193) and, thereby, attempting to establish the superiority of his SM statements compared to content published by mass media (for a detailed discussion about message framing on SM, please see Schmidt and Van Dellen 2022). This demonstrates that SM framing is also used to influence the adoption of values and more general interpretations of reality.

Taking a closer look at communication in SM reveals some differences compared to traditional media. Generally, leaders' digital and SM communication exhibits distinct characteristics (see also Kaye et al. 2022), including changes in information flows, the density of information, and the synchronicity of communication. Furthermore, there is a more frequent mixing of the actual message and other content (Voorveld 2019) as well as of sender and recipient (Chen, Diao, and Zhang 2011). While leaders tend to be the senders of information on traditional media, the fast feedback mechanisms of SM can also render them receivers of information and reactions almost simultaneously, making the 'relationship between subject and object in web communication merge into one' (Chen, Diao, and Zhang 2011, 6775).

In addition, SM communication tends to be more decentralised. By enabling everyone to participate in SM regardless of time and location, it becomes possible to communicate more easily and directly in the network without needing a main agent to lead the communication (Chen, Diao, and Zhang 2011). Using SM, leaders can bypass existing intermediaries, such as journalists or mass media, and communicate directly and immediately with their followers and stakeholders (Alghawi, Yan, and Wei 2014; Heavey et al. 2020; Huang and Yeo 2018).

Nevertheless, this also complicates the generation of a particular image or frame on SM for leaders, as any

image can be further developed or reinterpreted by any party, especially since all network participants can also communicate directly with each other. Adding to this complexity, especially on Twitter, communication is limited in space since posts may only contain a certain number of characters (Tur, Harstad, and Antonakis 2022). Therefore, leaders' communication has to take up much less space and messages must be brought across much more concisely.

These dynamics are further intensified by the fact that Twitter itself has been identified as a non-human leader (Poon and Kohlberger 2022). This is because, by its algorithms and platform structure, Twitter influences how conversations unfold, dictates narratives, and thereby shapes online communication. Therefore, when framing on SM, leaders must consider the characteristics of the communication channel (i.e. SM vs. non-SM, but also which SM-channel is used in particular, since these also differ significantly), which in turn leads to significant adjustments in framing and thus linguistic framing dimensions.

2.2. Framing as an instrument in leadership communication

As a key part of digital leadership, leaders' communication allows them to exert social influence by 'control[-ing] the context under which events are seen' (Fairhurst 2010, 2). Thereby, not only leaders' actions, such as public apologies after crises, can have a significant impact on firm perceptions and actions (Kim et al. 2015). Leaders' influence can also be more subtle, like using certain language attributes in public communication (Pan et al. 2018).

Accordingly, framing, being the process of 'select[-ing] some aspects of a perceived reality and mak[ing] them more salient in a communicating text, in such a way as to promote a particular problem definition, causal interpretation, moral evaluation, and/or treatment recommendation for the item described' (Entman 1993, 52) is also vital in the context of (digital) leadership and corporate communication. Thereby, leaders can use framing to adapt to the specifications of a certain communication context (i.e. SM).

Based on cognitive limitations inherent in every human being, the theory of framing assumes that all issues can be seen from different perspectives and interpreted in different ways (Chong and Druckman 2007). Consequently, how issues are framed by leaders has an influence on how these particular actions or information are understood (Chong and Druckman 2007; Entman 1993; Matthews et al. 2022; Tversky and Kahneman 1985).

When trying to analyse framing, four dimensions are especially important and frequently used in literature: Analytical Thinking, Clout, Authenticity, and Tone (Boyd and Pennebaker 2015). Analytical Thinking refers to the degree to which words are used that show formal thinking and logical mental processes (Pennebaker Conglomerates 2022), illustrating a rational and analytical way of framing. It further reveals a person's formal, and hierarchical patterns of thinking. Language scoring high in Analytical Thinking tends to be rewarded in academic settings and is correlated with more professional knowledge (Jordan et al. 2019). A low score in Analytical Thinking, in contrast, tends to be viewed as less cold and rigid, resulting in a friendlier, more intuitive, and personable perception of the person (Pennebaker et al. 2014).

Clout indicates the displayed status in society, leadership, and confidence (Kacewicz et al. 2014; Pennebaker Conglomerates 2022). A high score in Clout is therefore linked with leadership behaviour and dominance whereas lower Clout is linked with a more tentative but also a more self-centred communication (Fox and Royne Stafford 2021; Kacewicz et al. 2014).

Authenticity reflects the honesty of self-presentation and the degree of self-regulation (Pennebaker Conglomerates 2022). Individuals revealing a high score in Authenticity tend to use a more spontaneous and not self-regulated or filtered language and framing. Authenticity, therefore, refers to individuals who portray themselves in an authentic or honest way through their choice of language, making them more likely to be humble, personable, and vulnerable (Newman et al. 2003). Examples of this case are spontaneous conversations between close friends or political leaders with little to no social inhibitions. Texts that score low in Authenticity are mainly prepared texts (e.g. speeches that were written ahead of time) or texts that are characterised by formal writing.

Tone reflects the degree to which a person uses a positive or negative emotional tone (Pennebaker Conglomerates 2022) and therefore frames issues in a more positive or negative way. Announcing good news, such as the release of a new product, would thus be a text with a positive tone, whereas admitting a company's insolvency would be characterised by a negative tone.

2.3. Social media as a research context in leadership communication

From an academic perspective, although analysing SM is still a somewhat unconventional research method, it offers a fruitful way to study leadership communication that may have some advantages over standard research

(see also Bamberger and Pratt 2010; Strauss, Harr, and Pieper 2024; Tur, Harstad, and Antonakis 2022). First, SM offers the opportunity to examine leadership communication in vivo and to unravel long-term developments essential for understanding leaders' framing strategies unfolding over time. Here, especially through the context of SM as an interactive real-time medium, social influences and norms as well as the reference to the 'here and now' are particularly observable. Within the framework of SM analyses, interactions between, for example, corporate policy, personal interests of the CEOs, influence of the legal departments as well as interactions with non-organisational factors (i.e. war, economic crises) can be realistically represented, which could not be depicted with an experimental approach.

At the same time, 'specific rules, formal codes, and formats […] provide for a homogenous testing environment that limits extraneous influences and unmodeled variance' (Tur, Harstad, and Antonakis 2022, 5). Twitter posts, for example, are not allowed to exceed a certain length, file uploads are restricted to certain formats, and reactions can only be shown in predefined ways (e.g. likes, retweets). Therefore, all studied objects underly the same conditions (Tur, Harstad, and Antonakis 2022). Second, SM allows for detailed observation of framing because most networks that are used in professional contexts are primarily voice- or text-based and allow for an objective analysis of shared contents.

According to Matthews et al. (2022), leaders' SM usage is likely to advance and enlarge theoretical constructs all around leadership. Indeed, the widespread integration of digital technologies into various business domains and concepts (Larson and DeChurch 2020) has underscored the continuing need for practical and theoretical recalibration (e.g. Nambisan et al. 2017; Sebastian et al. 2020). Thereby, the exploitation of the extent to which traditional models of leadership can be generalised to digital contexts is crucial (Darioshi and Lahav 2021; Guerra-López and Blake 2011).

Especially when it comes to framing, the aforementioned specifications of SM, which allow the close and objective observation of leadership, simultaneously bring about the need to adapt certain aspects of leadership behaviour. Research has shown that several defining characteristics differentiate leaders' SM usage from other (analog) mediums of leader-follower exchange (e.g. face-to-face, public speeches, letters from the CEO): participants are geographically and temporally independent and completely mediated by technology (Raghuram et al. 2018).

SM platforms like Twitter are characterised by inherent technical principles such as low entry barriers and quick reaction times (Matthews et al. 2022), but also a character limit of posts (Tur, Harstad, and Antonakis 2022). Since framing describes the selection of particular aspects and presenting them more saliently in a communication context (Entman 1993), specifications given by SM-contexts must be addressed and leaders' adaptation to the communication medium is particularly relevant. Consequently, since the framework that SM creates for digital leadership differs essentially from non-digital leadership contexts (Avolio et al. 2014), we propose that leadership framing must also be adapted significantly.

With respect to the four linguistic framing dimensions mentioned above (Analytical Thinking, Clout, Authenticity, and Tone), we believe that they differ significantly, too. Based on the fundamental differences between SM and non-SM, we assume that all linguistic framing dimensions vary depending on the communication channel. Concerning each dimension, we expect the changes to be as follows:

First, leaders' Analytical Thinking should be higher in SM settings because of the style of microblog data: characteristics such as the character limit or the semantic ambiguity of text messages make it harder to infer emotions in written text (Daft and Lengel 1986; Hasan, Rundensteiner, and Agu 2019). Therefore, leadership statements would have to be more rational and thought through. Also, the faster and more uncontrollable spread of posts could lead to more caution and more precise wording (Badugu and Suhasini 2017). Short messages need more interpretation if they are not clearly formulated, so leaders are expected to use more Analytical Thinking to avoid room for interpretation. Drawing on an analysis of Chinese CEOs' SM communication, for instance, Alghawi, Yan, and Wei (2014) showed that followers favoured CEOs who adopted an expert role, which would imply exhibiting higher levels of Analytical Thinking, over those who positioned themselves as intuitive and empathic 'friends.' Therefore, we hypothesise the following:

> Hypothesis 1: Leaders show significantly higher levels of Analytical Thinking in SM than in non-SM surroundings.

Second, Clout should be lower in SM because leaders often engage in storytelling on SM within the context of personal branding (Eckardt et al. 2024; Mazzoni and Mincigrucci 2022). Personal branding encompasses practices to promote oneself across SM and non-SM platforms (Nolan 2015) to maintain visibility, influence, and a coherent digital identity (Chen, Rui, and Whinston 2021). Personal branding necessitates a focus on

oneself to present, for example, personal strengths, values, and success stories. To successfully establish a personal brand in SM contexts, leaders need to mobilise followers in a limited space (Banks et al. 2022). Therefore, they might want to diminish hierarchical distance and focus on approachability and closeness to followers. This is especially important for CEOs who are already seen as high-status individuals (Yang et al. 2023). By lowering their Clout, leaders can avoid the pitfalls of appearing impersonal and distant while, at the same time, emphasising their own personal narrative. This facilitates connecting with their audience on a more personal level than in traditional, usually more formal communications (e.g. letter to the investors, letter to the employees, financial reports). Hence, we hypothesise the following:

Hypothesis 2: Leaders show significantly lower levels of Clout in SM than in non-SM surroundings.

Third, leaders' Authenticity should be lower in SM because non-SM settings might invite for more spontaneity (e.g. in spoken interviews). Furthermore, the blending of company and leaders could result in CEOs tweeting in a fundamentally less authentic way, as they tweet vicariously for the entire company. Intentions and emotions in non-SM contexts can be supplemented in many ways such as facial expressions, gestures, and speech that can be observed by an audience. On the contrary, written expressions in SM environments differ significantly from non-SM contexts in a variety of characteristics (for an overview of the specifics of digital media, see Kaye et al. 2022).

A pivotal factor is the absence of para- and non-verbal cues (Duncan 1969; Kruger et al. 2005) – communication signals other than those conveyed by words (e.g. gestures, tones, and facial expressions) – necessitating the reliance on creative use of words for communicating intentions (Badugu and Suhasini 2017). Consequently, missing cues afford leaders to adapt their communication style (Walther 2011; Walther, Deandrea, and Tong 2010), which could be perceived as less authentic in contrast to their non-SM appearance. Furthermore, an authentic leader is defined to be perceived as 'confident, hopeful, optimistic, resilient, transparent, moral/ethical future-oriented, and [to] give […] priority to developing associates into leaders themselves' (Luthans and Avolio 2003, 243). However, SM platforms such as Twitter comprise numerous topics and emotional states (Hasan, Rundensteiner, and Agu 2019), rendering it challenging to manually create a comprehensive corpus of labelled aspects that covers all possible aspects of the concept of leader Authenticity. Accordingly, we hypothesise the following:

Hypothesis 3: Leaders show significantly lower levels of Authenticity in SM than in non-SM surroundings.

Fourth, we propose that leaders' Tone might be slightly more positive on SM. This is because, even if sharing information within a character limit might force leaders to focus on the information instead of the emotion surrounding this information (Hasan, Rundensteiner, and Agu 2019), it has been shown that SM is mostly used to share positive news (Yang and Liu 2017). In addition, Gardner et al. (2005) posited that a positive ethical climate would foster authentic relationships which is why it is especially critical to focus on good news and a rather positive Tone when communicating in a network that is characterised by interactions and a rapid spread of information (Heavey et al. 2020).

Maintaining a positive Tone in SM communications is also necessary because leaders frequently employ SM as a rather strategic tool for managing stakeholder impressions, especially when their personal or corporate images are threatened or compromised. In this vein, SM is also used as a form of immediate image repair (e.g. through apologies to stakeholders or as crisis responses; Hargie, Stapleton, and Tourish 2010; Mirbabaie et al. 2020).

Moreover, leaders often have sole control over their proprietary SM profiles, enabling them to operate outside of the corporate boundaries that typically govern traditional media. SM serve as unrestricted public environments where leader-follower interactions are openly displayed, influencing not only directly engaged individuals but also any observers (Barcelos, Dantas, and Sénécal 2018), which is why focusing on positive relationships is crucial. Consequently, we hypothesise that leaders' Tone tends to be slightly more positive in SM compared to non-SM contexts, as leaders aim to foster favourable impressions among a wider audience.

Hypothesis 4: Leaders show a significantly more positive Tone in SM than in non-SM surroundings.

3. Method

To answer our research questions, we first aimed at gaining a deeper insight into leaders' framing on SM in general. Accordingly, we gathered SM data from Twitter and analysed them with respect to the four linguistic framing dimensions (i.e. markers for Analytical Thinking, Clout, Authenticity, and Tone). In doing so, we omitted the 'ritualized use of questionnaires' (Fischer et al. 2020, 1) in leadership research and moved past traditional research templates (McDermott 2023) by employing digital technologies (i.e. Linguistic Inquiry and Word Count (LIWC) and topic modelling) to answer the call for rigorous qualitative research methods (Banks et al. 2023).

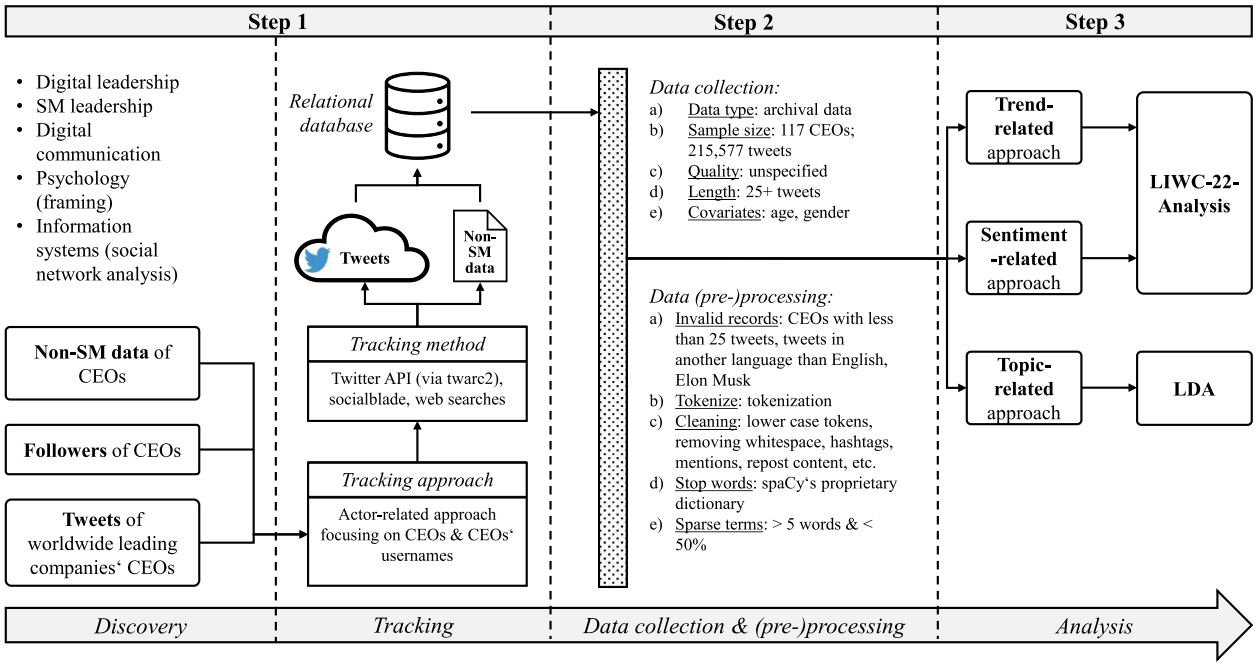


Figure 1. Methodological framework to conduct SM analytics in reference to Stieglitz et al. (2014, 2018) and Banks et al. (2022).

Second, we identified possible differences between Twitter and other public channels, while being informed by established methodological guidelines on SM data extraction, pre-processing, and analysis (e.g. Banks et al. 2018; Stieglitz et al. 2018). In detail, we compared our results with analyses of data from other, non-SM, sources. We therefore also provide a contingency perspective on CEOs' context-specific framing in different communication mediums (Cortellazzo, Bruni, and Zampieri 2019) – an endeavour that Johns (2024) acknowledges to constitute a deficit in prevalent leadership research.

As depicted in Figure 1, our methodological framework in reference to Stieglitz et al. (2014; 2018) and Banks et al. (2018) encompasses three main steps: (1) discovery and tracking, (2) data collection and (pre-)processing, and (3) data analysis. These steps are elaborated in the following sections.

3.1. Discovery and tracking

As stated by Banks et al. (2022), researching digital leadership phenomena affords an interdisciplinary perspective on the behaviours of leaders. In our study, we focus on an interdisciplinary contingency perspective on context-specific (see also Johns 2024) SM framing of worldwide leading companies' CEOs in comparison to framing in traditional communication realms.

Drawing on the research domain of information systems, which provides a sophisticated methodological vehicle for social network analysis (Stieglitz et al. 2014, 2018), we integrate into our interdisciplinary perspective digital and SM leadership (e.g. Banks et al. 2022; Matthews, Wang, and Kelemen 2024; Tur, Harstad, and Antonakis 2022), social science focused on digital communication (e.g. Bakardjieva, Felt, and Dumitrica 2018; Strauss, Harr, and Pieper 2024), and psychology focused on SM framing (e.g. Entman 1993; Hansson, Fuoli, and Page 2024; Leong 2024).

To answer the RQs and shed light on CEOs' context-specific framing (i.e. CEO's framing in SM vs. non-SM environments), we follow an actor-related tracking approach focusing on both, CEOs' non-SM data as well as CEOs' tweets from Twitter, to uncover 'latent structures and patterns' (Chinnov et al. 2015, 2) in their digital communications. Following previous leadership research (e.g. Aktas et al. 2016; Bligh, Kohles, and Meindl 2004; Craig and Amernic 2018; Harrison et al. 2019, 2020; Malhotra and Harrison 2022), we included non-SM data, such as transcripts of interviews, public speeches, letters from the CEO, publications on websites, and messages to the employees for all leaders.

Regarding SM-data, we utilised Twitter, as it is an established SM platform that allows users to share text-based information (however, the integration of links, photos etc. is allowed) to a (subscribing) audience, so-called followers. According to Chen, Rui, and Whinston (2021, 500), leaders personal Twitter accounts are 'a natural tool to project an image in the minds of others.' Indeed, over the years, Twitter has become a popular platform not only for personal branding of leaders but also for researchers (for an overview see

Matthews et al. 2022), due to its high standardisation as well as its availability (Heavey et al. 2020). Tweets are allowed to be up to 280 characters long (until 2017, only 140 characters were allowed) and can be accessed by followers or any Twitter user that searches for either the profile or a word/hashtag that was used (Tur, Harstad, and Antonakis 2022). In addition, we enriched our tracked data with additional metadata, such as follower counts, the CEO's gender, and the CEO's age. Due to the vast amount of data, we used a relational database to store them. This allowed us to perform efficient search and filter queries.

3.2. Data collection and (pre-)processing

The data for our study were collected from the CEOs of the world's leading companies, i.e. companies listed in the Fortune 500 Ranking, the S&P 500, or the Nasdaq-100. We gathered the usernames of all CEOs' Twitter accounts, resulting in 117 users in total.

We then determined every leader's number of followers to get an overview of their estimated audience-size. Additionally, we collected all tweets from each leader, from the first post in August 2008 until the end of our data-collection in November 2022, leading to a total of 215,577 tweets. The Twitter application programming interface (API) was accessed using 'twarc2.'[2]

Following Marouf, Rokne, and Alhajj (2022), we excluded mentions (@) and hashtags (#). We also deleted links, symbols and emojis since they do not provide any information for text analysis. Additionally, we deleted posts that consisted only of mentions, hashtags and/or links. However, unlike Marouf, Rokne, and Alhajj (2022), we did not delete reposts since many users also added own text when sharing posts from other users. Therefore, we deleted the content of the repost itself but kept the text of the person sharing the respective tweet.

Following the methodological approach of Golder and Macy (2011), we then excluded users with less than 25 tweets as well as users that did not tweet in English. To avoid biases, we also excluded Elon Musk since he is the owner of Twitter and several studies have addressed his unique appearance on this SM channel (e.g. Ante 2023; Browne 2021; Seigner et al. 2023). Furthermore, he differed significantly from the rest of the group (i.e. he had ten times more followers than the second most famous CEO in the dataset and more than 15,000 posts; retweets excluded). Our final sample consisted of 88 users (i.e. CEOs) and 80,210 tweets in total.

In a second step, we aimed to obtain further insights into changes of linguistic framing over time. Thus, we enriched the data set with the date on which a specific tweet was posted. Based on these dates, we calculated weekly averages for the four dimensions. Furthermore, we added data on the age of each executive. In most cases, birth dates were publicly reported, so we were able to calculate the CEOs' age at the time of data collection. Based on this, we also calculated the age of the CEO when each tweet was published. Some CEOs did not provide their birthday or month online (e.g. on BoardEx), so we had to limit ourselves to the year. To approximate age in these cases, we assumed that leaders born before June 1st were the older of the two possible ages, and all leaders born after that date were the younger. For leaders who reported only their birth year, we assumed they obtained the younger age.

Additionally, we added the number of followers at the respective time. Therefore, we used an online tool called 'socialblade'[3] which allows collecting follower information on a daily basis. However, since socialblade starts collecting daily followers only after a username has been searched at least once, not all follower details could be assessed. For 37 leaders, there was no information available since we were the first to search for their usernames. For 51 leaders, however, we could assess more than two years of follower data which, due to the high posting frequency on this network, is a sufficient timespan for a platform like Twitter to provide longitudinal data. For these 51 leaders, we then calculated weekly averages for the number of followers, resulting in 8,001 weekly averages.

To enrich the data gathered, we also collected additional, non-SM, data from each CEO. In further detail, we obtained transcripts of interviews (62), public speeches (58), letters from the CEO (41), publications on websites (18), and messages to the employees (12) for all leaders (upon availability), leading to 191 additional documents with between 121 and 9,457 words per transcript (354,339 words in total). As part of this, we ensured that the content of all non-SM sources also came from the respective leader. This was done by one author who removed all parts of communication where it was obvious that people other than the leader were speaking or writing. For example, interviewer questions and comments were deleted from interviews to make sure that only leaders' content was analysed. However, this only led to the deletion of document sections, not entire documents. All non-SM sources were identified through individual web searches for each CEO. In addition, video platforms were searched for public speeches and financial statements were manually analysed for publications by the CEOs.

As a last step, we wanted to gain further insights on words CEOs used frequently to deepen our understanding of linguistic leadership framing in SM and non-SM contexts. Consequently, we conducted a topic analysis to see which topics CEOs were talking about most. Before running Latent Dirichlet Allocation (LDA), however, several data preparation and (pre-)processing steps are necessary.

Initially we conducted standard data (pre-)processing steps following Banks et al.'s (2018) recommendations. Drawing on established Python libraries, such as spaCy 2.0[4] and Gensim[5] (Rehurek and Sojka 2011), we removed sparse terms, such as words occurring in less than five tweets as well as in more than 50% of the documents (Ghanoum 2021). Following Banks et al. (2018), tweets and non-SM documents were reduced to single words within tokenization. Certain tokens such as those for adverbs, conjunctions, numbers, etc. were removed and an ID was assigned to each word from a tweet.

Furthermore, we cleaned the data by converting all text to lowercase, removing punctuation, mentions, hashtags, links, numbers, and symbols. Following this, we removed stop words – common, high-frequency, non-informative words that hinder inference of meaningful topics (Banks et al. 2018; Blei, Ng, and Jordan 2003; Schofield, Magnusson, and Mimno 2017). For instance, common words, such as 'you,' 'a,' or 'and,' were removed to reduce noise, since they convey little meaning (Tonidandel et al. 2022). Following the argument by Tonidandel et al. (2022), we opted not to employ stemming or lemmatisation in our LDA model despite recommendations by Banks et al. (2018) due to 'lexical ambiguity' (Laureate, Buntine, and Linger 2023, 14245), mixed evidence regarding their efficacy in large datasets, and potential negative impacts on topic modelling performance (Hickman et al. 2022; Schofield and Mimno 2016; Schofield, Magnusson, and Mimno 2017).

Finally, we calculated the semantic coherence of topics utilising the Coherence Value (C_V) metric for each context as outlined by Röder, Both, and Hinneburg (2015), to ascertain the optimal number of topics (for an overview, see Appendix A). The rationale behind using C_V is based on its high correlation with human judgment (Röder, Both, and Hinneburg 2015) and its established utilisation in research for evaluating topic clusters (e.g. Khan and Ali 2023; Kraus et al. 2023; Wahid et al. 2022). Consequently, the nine most significant SM topics ($C_V = .4379$) and the seven most significant non-SM topics ($C_V = .4552$) were determined, with each tweet/document being categorised under one topic (for an overview, see Appendix B).

3.3. Data analysis

For data analysis, we adopted three approaches: a trend-related, a sentiment-related, and a topic-related approach. Regarding the former two, we analysed the data using LIWC-22, a text-analysis tool that has been proven to be specific, sensitive, reliable, and valid (Bantum and Owen 2009; Boyd and Pennebaker 2015; Pennebaker and Francis 1996). LIWC has been used in multiple different studies in various fields (e.g. Drouin et al. 2017; Fox and Royne Stafford 2021; Jordan et al. 2019; Obschonka, Fisch, and Boyd 2017). Especially in leadership research, LIWC has proven to be a valuable tool to analyse leaders' language as a type of unobtrusive measure (Matthews, Wang, and Kelemen 2024).

LIWC-22 counts and analyses words in a given text and matches them to predefined dictionaries. The programme calculates the results in two different ways. In the first case, the percentage of a dimension in the total number of words in each text is reported. Consequently, a value of 1.7 for 'you' would mean that, on average, the word 'you' appears in 1.7% of the tweet. The dimensions we mainly deal with belong to the second group of results, the so-called summary measures. There are four summary measures in total: Analytical Thinking, Clout, Authenticity, and Tone. Although LIWC-22 includes both positive tone and negative tone dimensions, the Tone variable puts the two dimensions into a single summarising variable considering whether the text expresses an overall positive or negative emotional tone (Cohn, Mehl, and Pennebaker 2004).

The scales from all summary measures range from 1 (minimum) to 99 (maximum) and are standardised. For Tone, numbers below 50 indicate a more negative Tone, while numbers above 50 indicate a more positive Tone (Cohn, Mehl, and Pennebaker 2004). Tausczik and Pennebaker (2010) (see also Bantum and Owen 2009; Seih, Beier, and Pennebaker 2017) describe extensive efforts to ensure the validity of LIWC-22's classification of words into the appropriate correct sub-dictionaries, rendering LIWC-22 a solid and well-proven tool for text analysis.

Regarding the latter approach (i.e. the topic-related approach), topic analysis was performed by topic modelling with LDA using Natural Language Processing (Blei, Ng, and Jordan 2003; Xue et al. 2020). LDA, a hierarchical Bayesian model, identifies hidden topics within documents by assuming each document is a mixture of a limited number of topics and that each word's presence is attributable to one of these topics (Blei, Ng, and Jordan 2003; Jelodar et al. 2019). Consequently, LDA not only reduces data dimensionality from

thousands of possible word frequencies in tweets and non-SM data to a manageable number of topics but also assigns a mixture of these topics to each document (Blei, Ng, and Jordan 2003; Zhou et al. 2024).

This process is quantified in the document-topic matrix, where each document is represented as a probability distribution over all topics (Banks et al. 2018). Hence, LDA is well-suited for our research as it effectively uncovers latent textual structures within large datasets, facilitating the extraction of underlying themes related to CEOs framing in SM and non-SM contexts without reliance on pre-defined categories (e.g. Barnes et al. 2021; Jelodar et al. 2019; Zhou et al. 2024).

Coding was primarily done in a Jupyter Notebook using Python 3.9 as programming language. For statistical calculations, we used SPSS version 29. Accounting for the hierarchical structure of the data, we calculated averages per leader. Due to high standard deviations (SD) in framing levels, likes, shares, and other followers' reactions, resulting in possible outliers in all dimensions, the mean as well as the median were calculated and analysed. Additionally, we accounted for the non-normal distribution of some variables by using statistical methods that did not assume normal distribution in those cases.

4. Results

The 88 CEOs consisted of 78 men (88.6%) and 10 women (11.4%), which is higher than the average for women in the Fortune 500 (8.2%), the S&P 500 (6.4%), or the Nasdaq-100 (7%) indexed companies. Leaders were on average 56.58 years old (see Table 1). On Twitter, they had between 213 and 13,721,598 followers and posted between 28 and 9,148 tweets with 343.73 likes on average (SD = 1,467.93).

On average, all leaders show relatively high levels of Analytical Thinking (75.89; median: 77.92) a relatively high Clout (68.02; median: 68.63), low Authenticity (35.78; median: 35.57), and a positive Tone (74.24; median: 75.40). However, all leaders reveal high SDs (on average: Analytical Thinking: 27.11; Clout: 30.05; Authenticity: 35.14; Tone: 32.68), illustrating high intraindividual differences. Thus, across posts, a leader has a wide range within the different dimensions and varies framing. An exemplary illustration of intraindividual changes can be seen in Figure 2.

Between leaders, however, SDs of dimension averages are much lower with 8.70 for Analytical Thinking, 10.21 for Clout, 6.03 for Authenticity, and 10.80 for Tone, displaying low degrees of interindividual differences. This in turn shows a similar framing across all CEOs; despite intraindividual variations, all leaders settle at a very similar framing level.

Comparing gender averages, we did not find any significant differences; neither in the four dimensions, nor in average likes, quotes, replies, retweets, or in the word count. When observing weekly averages, the job position as well as the time a tweet was posted also had no significant effect on framing behaviour, indicating that there were no significant changes in linguistic framing over time.

Concerning additional factors, a higher word count was correlated with higher levels of Analytical Thinking (.35; $p < .01$; median not significant), higher Clout (.73; $p < .01$; median: .70; $p < .01$), and a more positive Tone (.50; $p < .01$; median: .32; $p < .01$). In addition, Analytical Thinking (−.31; $p < .01$; median not significant), Clout (−.34; $p < .01$; median: −.35; $p < .01$), and Tone (−.32; $p < .01$; median not significant) were negatively correlated with a higher tweet count. The number of followers as well as the age of the CEO did not correlate or have significant ($p < .01$) effects on any of the four dimensions.

Analysing additional transcripts from non-SM sources revealed significantly ($p < .01$) lower levels of Analytical Thinking (61.19) than in SM surroundings (see Table 2). Furthermore, leaders showed significantly higher ($p < .01$) Clout (85.23) and Authenticity (41.90) in non-SM contexts. Furthermore, they showed a significantly more negative Tone (67.96; $p = .01$). Medians showed significant ($p < .01$) differences as well. Therefore, as summarised in Table 2, all our hypotheses can be supported.

Since the two sources differ significantly in their wordcount, we also conducted a regression analysis for the average word count and the averages in the four dimensions for both SM and non-SM to see in how far the wordcount might have affected the dimensions. The analysis showed significant ($p < .01$) results for all dimensions (Analytical Thinking: .41, median: .51; Clout: .36, median: .20; Tone: .35, median: .43; Authenticity: .34, median: .49), underlining the importance of the contextual restrictions of Twitter.

Table 1. Descriptive statistics.

	Mean	SD	Median	Min/Max	Skewness
Age	56.58	5.73	57.29	42.92/71.58	−.37
Followers	348,717.82	1,663,965.35	8038	213/13,721,598	.61
Posts	911.48	1438.84	366	28/9,148	1.14
Average likes per tweet	343.73	1,467.93	33.09	.6/11,143.71	.63

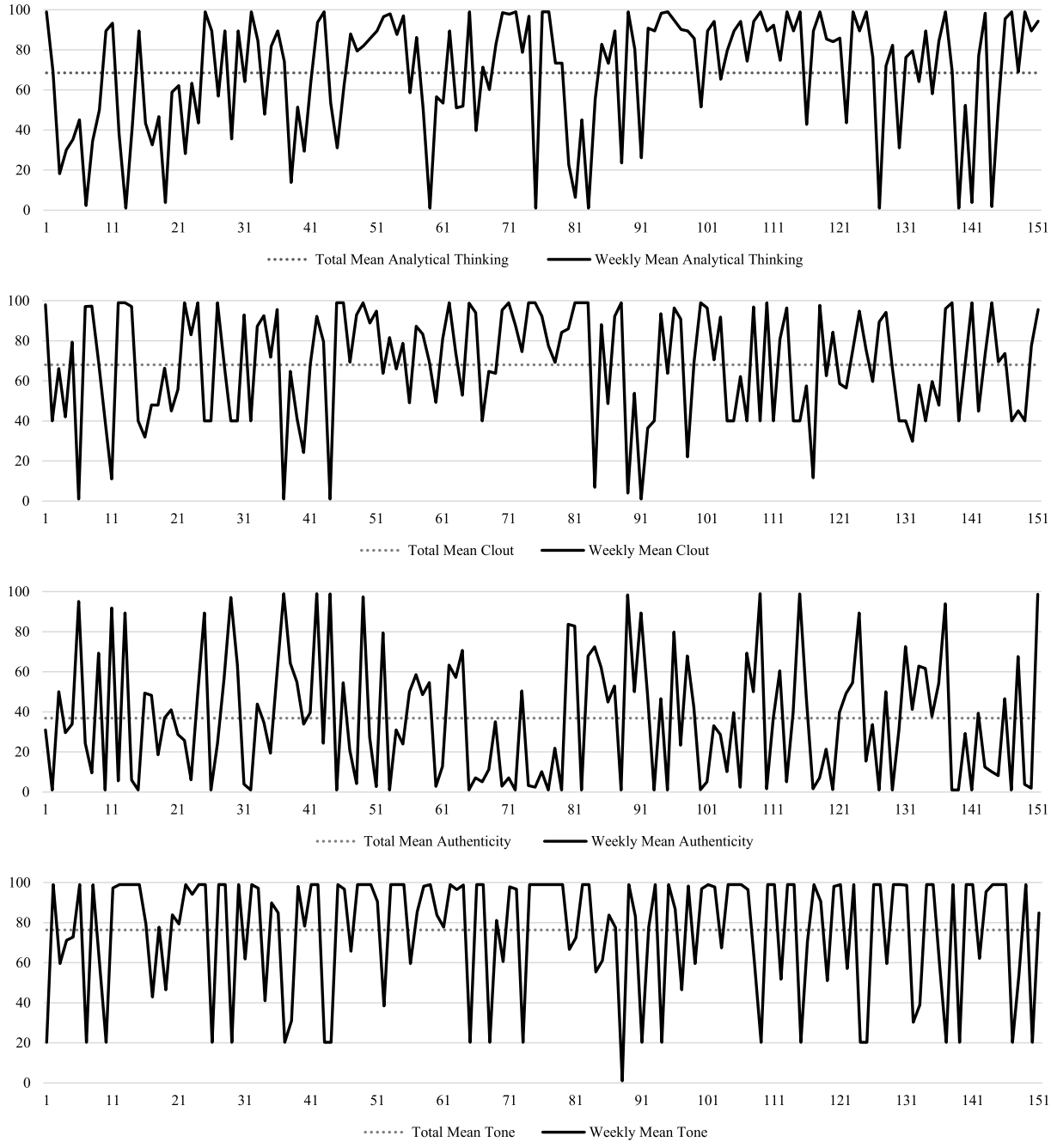


Figure 2. Exemplary intraindividual differences in framing over time.

Table 2. Levels of analytical thinking, clout, authenticity, and tone.

	Analytical Thinking		Clout		Authenticity		Tone	
	Mean	*SD*	*Mean*	*SD*	*Mean*	*SD*	*Mean*	*SD*
Interview *(non-SM)*	44.58	19.94	79.27	20.20	47.75	17.71	57.77	16.99
Public speech *(non-SM)*	62.51	18.58	81.58	18.77	46.67	17.12	66.40	16.03
Letter from the CEO *(non-SM)*	76.69	12.49	93.32	9.09	30.28	12.56	82.73	11.69
Publication *(non-SM)*	71.00	18.36	82.01	11.15	41.26	20.43	62.43	15.69
Message to employees *(non-SM)*	64.78	15.53	91.64	11.28	39.29	12.00	67.90	26.08
Total mean *(non-SM)*	61.19	17.36	85.23	13.82	41.90	13.98	67.96	15.53
Twitter *(SM)*	75.89	8.70	68.02	10.21	35.78	6.03	74.24	10.80

Regarding the topic analysis of both SM and non-SM contexts, LDA revealed nine main topics in SM and seven main topics in non-SM contexts for CEOs based on semantic coherence (for an interactive visualisation of topics please see the supplementary materials in Appendix C; for a complete overview of all identified topics, please see Appendix B). The most dominant SM-topic only contains 13,887 tweets (17.31%), however, more than one third of all non-SM documents (37.17%) are categorised under a single topic. Noticeably less tweets/documents (between 5,321 (6.68%) and 10,474 (13.06%) tweets and between five (2.62%) and 40 (20.94%) non-SM documents per topic) were assigned to the other identified topics (for an overview, please see Table 3).

In SM contexts, CEOs focused on promoting positive sentiment, achievements, and customer relationships. Taking a closer look at the top five topics in SM contexts (see Table 3), the most prevalent topic (i.e. T1) contained words such as 'good' and 'love,' which reflects a communication strategy centred around personal engagement. The frequent use of words like 'read,' 'time,' 'think,' and 'know' (T1) also suggests a more thoughtful and conversational tone, aiming to create a sense of connection and thought leadership. Hence, CEOs fostered a relatable and approachable persona, emphasising a lower Clout as they engage directly with their audience, often discussing broader life experiences or encouraging followers to reflect alongside them.

The frequent use of words like 'welcome,' 'live,' 'get,' or 'join' (T1, T5) further points to an interactive approach and indicates lower Clout, aiming to foster a sense of community/team engagement (e.g. 'community' in T7; 'team' in T2) and involvement among followers. As surfaced by LDA, CEOs' lexical choices indicate a dynamic, personalised communication style as well as an attempt to humanise the corporate persona and directly connect with their followers.

This is also evidenced by CEO's frequent use of optimistic and positive words (i.e. higher Tone), such as 'good,' 'love,' 'great,' 'proud,' 'honor,' 'excited,' or 'thank' (T1, T2, T3, T5). Interestingly, CEOs utilised SM to communicate corporate culture and personal engagement, as they shared their appreciation towards work teams and collaborations (e.g. T2) or celebrated their own leadership achievements (T3). Regarding the former, T2 is characterised by a strong focus on team and corporate culture, as indicated by the frequent use of 'team,' 'thank,' 'great,' and 'congrats.' Regarding the latter, T3 is characterised by a strong focus on celebrating leadership achievements and company values, as represented by frequent words such as 'work,' 'proud,' 'leader,' 'honor,' 'congratulation,' and 'thank.'

Overall, this suggests that CEOs used Twitter to publicly acknowledge and celebrate achievements and milestones, fostering a positive work environment and showcasing their appreciating and recognising leadership style. Thereby, CEOs not only enhanced employee morale but also effectively conveyed their personal and employer brand, for instance, as an exemplary workplace where (team) performance is publicly valued (see also Venciute, April Yue, and Thelen 2024).

In this vein, CEOs also utilised SM to represent their corporation as forward-thinking, customer-driven, and technologically adept within the wider industry landscape: terms such as 'customer,' 'business,' 'technology,' and 'innovation' (T4) signals that CEOs also framed their communication around the dual imperatives of addressing both, customer needs (e.g. 'customer,' 'partner,' or 'help' (T4)) and digital transformation (e.g. 'digital,' 'innovation,' 'new,' or 'technology' (T4)). Furthermore, the inclusion of terms such as 'partner' and 'drive' (T4) indicates the collaborative nature of corporations through strategic partnerships to meet dynamically changing market demands.

Regarding CEOs communications in non-SM contexts, CEOs emphasise structured, strategic communication, often focusing on long-term planning, financial growth, operational details, and corporate governance. Analysing the top five topics in non-SM contexts (see Table 3) reveals CEOs engagement with overarching topics that are pivotal for corporate governance, social responsibility, and technological advancements.

In line with Fedorova, Nevredinov, and Drogovoz (2024, 694), who applied LDA to CEOs' interviews and showed that the 'coronavirus pandemic was […] of great importance,' the most prevalent topic (T1) is themed around the 'impact' (T1) of the coronavirus pandemic (e.g. 'pandemic,' 'health,' 'crisis,' or 'patient' (T1)) on the 'organization' (T1) and 'community' (T1) as well as on the CEO's (and/or the corporation's) role in the global pandemic (e.g. 'ensure' or 'goal' (T1)).

Additionally, terms such as 'cloud,' 'platform,' 'digital,' 'network,' and 'security' (T3) indicate, that CEOs utilised non-SM communications to discuss the technical infrastructure of their corporation as well as their approach towards digital transformation. Unlike SM contexts, where communication may be more engaging and accessible, non-SM channels allow for more in-depth and specialised discussions about critical issues such as cybersecurity, data management, and the strategic importance of digital platforms for the corporations' digital transformation initiatives.

This is also reflected in the different target groups: the rather specific words (e.g. 'cloud,' 'platform,' 'system,' or 'security' (T3, T5)) indicate a more informative

Table 3. CEOs top five topics in SM and non-SM contexts.

	SM context			Non-SM context		
T	No. of tweets	% of corpus	Top ten tokens per topic	No. of docs	% of corpus	Top ten tokens per topic
T1	13,887	17.31%	.038*'good' + .029*'love' + .025*'read' + .025*'think' + .021*'time' + .020*'know' + .019*'thing' + .017*'okta' + .015*'get' + .015*'want'	71	37.17%	.014*'community' + .012*'health' + .009* 'culture' + .009*'impact' + .007*'pandemic' + .007*'ensure' + .007*'goal' + .007*'organization' + .007*'patient' + .007*'crisis'
T2	10,474	13.06%	.156*'great' + .122*'thank' + .063*'team' + .019*'time' + .018*'enjoy' + .017*'customer' + .016*'talk' + .015*'congrat' + .015*'dell' + .015*'today'	40	20.94%	.015*'get' + .013*'lot' + .012*'talk' + .012*'say' + .010*'life' + .010*'like' + .009*'different' + .008*'let' + .008*'happen' + .008*'idea'
T3	9,964	12.42%	.030*'work' + .024*'proud' + .022*'honor' + .022*'world' + .019*'thank' + .019*'year' + .018*'congratulation' + .016*'leader' + .016*'team' + .014*'company'	30	15.71%	.024*'cloud' + .019*'platform' + .016*'digital' + .014*'datum' + .013*'security' + .011*'network' + .008*'organization' + .008*'partner' + .008*'device' + .008*'application'
T4	9,634	12.01%	.024*'customer' + .022*'business' + .019*'new' + .019*'technology' + .017*'help' + .017*'drive' + .016*'innovation' + .014*'industry' + .013*'digital' + .013*'partner'	24	12.57%	.021*'growth' + .020*'quarter' + .017*'strong' + .014*'revenue' + .013*'increase' + .011*'client' + .010*'deliver' + .009*'expect' + .009*'sale' + .009*'term'
T5	8,220	10.25%	.076*'look' + .027*'join' + .027*'welcome' + .025*'week' + .023*'live' + .020*'today' + .019*'event' + .018*'excited' + .017*'congrats' + .016*'new'	16	8.38%	.043*'energy' + .019*'power' + .019*'system' + .015*'emission' + .014*'carbon' + .014*'water' + .013*'vehicle' + .012*'gas' + .011*'investment' + .011*'electric'
∑	**52,179**	**65.05%**		**181**	**94.76%**	

Note: T = Topic; Percentages are rounded to two decimals.

communication style that reflects an audience with technical and business knowledge (i.e. stakeholders such as investors, regulators, or business partners). Not surprisingly, CEOs also heavily talked about corporate financial performance and growth, as highlighted by terms such as 'growth,' 'quarter,' 'revenue,' or 'increase' (T4).

Accordingly, CEOs framed their communications around quantitative metrics and financial data to emphasise to stakeholders, to what extent the corporation met or exceeded market expectations. However, besides rather formal fiscal data, LDA also revealed that CEOs communications focused on energy and sustainability issues (e.g. corporate social responsibility).

In comparison to the SM context (see Appendix B), however, there is a clear divergence in CEOs framing: While both platforms address sustainability, SM communications tend to link sustainability with broader positive messages (e.g. support for community or innovations in clean energy (T7)), while non-SM communications presented it in more technical and policy-focused terms (e.g. 'energy,' 'system,' 'emission,' 'carbon,' 'investment,' or 'power' (T5)). Hence, CEOs delved deeper into the specifics of environmental 'investments,' 'energy systems,' and regulatory compliance by framing more precise, with a focus on conveying the corporations' commitments to reducing emissions, investing in clean energy technologies, and aligning with sustainability standards.

5. Discussion

In our study, we examined the manifestation of leaders' framing in SM vs. non-SM contexts. The distinct characteristics of SM (e.g. communication speed, character limits, and the absence of non- and paraverbal cues) necessitate fundamental changes in SM communication compared to non-SM contexts. In particular, the findings of our study suggest that leaders' linguistic social media framing is characterised by high Analytical Thinking, rather high (but still lower than in non-SM contexts) Clout, low Authenticity, and a rather positive Tone.

Collectively, these dimensions form a distinct SM behaviour, where each dimension influences the others. Higher Analytical Thinking, for instance, can lead to more prepared language, and, consequently, to lower Authenticity. We found that, on average, there is consistent linguistic SM framing among CEOs – across time as well as across leaders – that significantly differs from non-SM contexts. However, linguistic framing dimensions vary greatly within the individual leader.

We contribute to existing literature on digital as well as SM leadership (e.g. Avolio, Kahai, and Dodge 2001; Heavey et al. 2020; Matthews et al. 2022; Tigre, Henriques, and Curado 2025) in multiple ways. First, we reveal a contextuality of SM framing, thereby overcoming parts of the acknowledged context deficit in leadership research (Johns 2024). Twitter, while allowing direct interaction and personalisation (Chen, Diao, and Zhang 2011), also restricts leaders' communication and confronts them with higher contextual expectations (Blankespoor 2018). The adaptation to these constraints imposed by the social medium (Daft and Lengel 1986) results in specific SM framing that enables adequate communication of the content to be shared.

Second, we show that these differences between the two contexts result from their distinct characteristics rather than from personal traits of each leader since these manifestations of SM framing appear independently of the person. Consequently, in line with Poon and Kohlberger (2022), our study provides strong evidence that the medium (i.e. Twitter) significantly shapes leadership communication and behaviour, challenging the common, yet increasingly questioned, notion that leadership behaviour is purely individual and based on the personality of the leader (i.e. Northouse 2013; Podsakoff, MacKenzie, and Bommer 1996). There are various possible reasons for the differences in each dimension of Analytical Thinking, Clout, Authenticity and Tone, which we discuss next.

In SM contexts, spatial limitations make it necessary to think especially carefully before tweeting to avoid misunderstandings or even defamatory reactions from followers that might directly go viral (Etter, Ravasi, and Colleoni 2017; Wang, Reger, and Pfarrer 2019). Therefore, leaders show higher degrees of Analytical Thinking. In non-SM contexts, however, leaders might rather want to transmit emotions to their stakeholders and therefore use more intuitive and honest, or less prepared, language.

In the case of Clout, leaders could assume that dominance and the explicit expression of leadership can also be misunderstood in digital contexts. This is because some aspects of communication, such as the tone of the voice, cannot underline what has been said (Daft and Lengel 1986).

In addition, lower degrees of leaders' Authenticity in SM contexts corroborate prior research findings (Capriotti and Ruesja 2018) that CEOs have a lower interaction rate on Twitter and rather use it to share their personal ideas and information. In real life, especially in longer written statements, leaders might want to appear more approachable, humble, and personable. In many non-digital contexts, the presence or

a statement from a CEO is especially important when followers or stakeholders have to be addressed and/or motivated. Therefore, showing Authenticity and closeness to the audience has been proven to be most efficient (Guerrero, Lapalme, and Séguin 2015; Men and Stacks 2014; Walumbwa, Christensen, and Hailey 2011).

Lastly, leaders' more positive Tone in SM contexts can be supported by research showing that leaders tend to share only positive news on SM (Yang and Liu 2017). When sharing irrelevant or daily information, which is often the case on Twitter as a microblogging service (Huang and Yeo 2018), a positive Tone might also be more appropriate. Furthermore, in SM, a more positive Tone can also help to avoid negative reactions such as 'shitstorms', which can cause lasting damage to their reputation (Etter, Ravasi, and Colleoni 2017; Wu et al. 2022).

However, we found significant changes in leadership framing when more characters could be used to express oneself. This is another reason which supports our statement that the context of SM, especially Twitter, forces leaders to adapt their framing to be equally efficient using less space and having other possibilities of expressing and/or correcting themselves than in real life.

These findings are in line with prior research on digital leadership (e.g. Avolio et al. 2014; Cortellazzo, Bruni, and Zampieri 2019; Tigre, Henriques, and Curado 2025). Drawing on the lens of contingency theory, it has been argued that 'leadership should adopt a style and behaviors that match the context' (Cortellazzo, Bruni, and Zampieri 2019, 4). For instance, our topic analysis revealed that sustainability issues were framed differently in SM and non-SM contexts: on Twitter, sustainability was often framed in relation to positive engagement and broad social appeal (e.g. 'support,' 'community,' 'change,' 'future,' or 'hope'), whereas in non-SM communication, it was addressed through more technical and policy-oriented language, with a focus on energy systems and emission reduction (e.g. 'carbon,' 'investment,' or 'power'). Thus, a distinct and consistent framing strategy can serve the purpose of addressing contextual specifications (Blankespoor 2018) and being able to pursue digital leadership in SM, as leaders must adapt their messages to the medium and audience (e.g. Strauss, Harr, and Pieper 2024).

In summary, in line with further research that has emphasised the necessity for distinct digital models (Avolio et al. 2014), we identify distinct SM framing that constitutes a certain consistency and otherness to real-life communication contexts. While SM communication tends to focus on fostering engagement, positivity, and immediacy, non-SM communication allows for more depth, technical detail, and strategic messaging. This aligns with our findings on leadership framing strategies, where topics such as digital transformation and financial performance were framed in highly technical and data-driven ways in non-SM channels. In contrast, SM discussions of innovation and growth were more likely to emphasise accessibility and excitement, aimed at engaging a broader audience with simpler, more emotional language.

Lastly, we reveal that leaders' intraindividual framing tactics change over time. Our longitudinal data show an inconsistent framing within the individual leader. This might be explained by our finding that they mostly tweet about topics of public interest (e.g. see Appendix B). At the same time, this focus on topicality provides an explanation for the coexistence of intraindividual variance and interpersonal consistency: on the one hand, leaders have to adapt their framing to Twitter as a general communication context as well as its social norms (Cortellazzo, Bruni, and Zampieri 2019).

At the same time, however, they must also account for its real-time characteristics and pay attention to specific topics within this broad spectrum that might attract public interest at the time of the tweet. In other words, a crisis must be framed differently than an Initial Public Offering (IPO). Our topic analysis underscores that crisis framing on Twitter often highlights urgency and community resilience, while non-SM communications are more likely to focus on long-term impact and strategic mitigation. Therefore, leaders' linguistic framing also has to be characterised by high degrees of flexibility, since a wide range of very different topics is covered on Twitter (Hasan, Rundensteiner, and Agu 2019).

Our study also has some significant implications for practice. For corporations, our findings demonstrate a clear imperative to cultivate distinct communication strategies tailored for both SM and non-SM contexts. A leader's offline persona can diverge considerably from their online persona. This is a critical factor demanding careful consideration in corporate communication planning. Consequently, specialised training programmes for leaders' SM communication are becoming a necessity to appropriately convey both the personal and corporate brand. Such training programmes are essential because SM platforms such as Twitter require 'a different skill set and a different sensibility than other forms of marketing and communications' (Malhotra and Malhotra 2016, 78) and could equip CEOs with the necessary skills to navigate SM platforms effectively, develop optimal engagement strategies, and balance the inherent risks with the authentic expression of their individual personality. Particularly

during crises, a leader's ability to act with composure on SM is crucial for leveraging its potential to influence public perception and maintain stakeholder trust. For instance, genuine apologies from leaders on SM can effectively mitigate the adverse impacts of crises (e.g. Kim et al. 2015; Malhotra and Malhotra 2016) and reshape the narrative among followers (e.g. Mirbabaie et al. 2021).

For communication professionals, including SM managers and public relation experts, our study highlights the need to comprehend the platform-specific characteristics of digital communication. Our results illustrate, for instance, that the character limits on certain platforms can profoundly impact communication style. This limitation is likely also mirrored in follower expectations, requiring a focused approach to posting within these parameters. Accordingly, communication professionals must thoughtfully select appropriate SM platforms and carefully consider their technological affordances in relation to corporate goals (see also Schmidt and Van Dellen 2022). However, this is a challenging task due to the dynamic and contingent nature of both the corporate context and the continual evolution of SM platforms.

Furthermore, we encourage the intentional leveraging of modern analytical technologies to assess performance of posts. Our research demonstrates the feasibility of moving beyond rudimentary success metrics provided by platforms, such as likes and reposts, to conduct deeper analyses of communication, for example, by examining their framing dimensions. Such in-depth analysis can provide insights into which types of posts are more or less likely to generate attention, get retweets, or achieve their intended communicative objectives.

Finally, our work carries significant implications for future leaders. Our findings underscore the growing demand for distinct SM communication skills. Therefore, even if there are no trainings provided by the company, leaders should proactively invest in understanding and engaging with SM, as its skilful utilisation holds immense potential for demonstrating effective leadership.

We also encourage leaders to embrace adaptability; a critical skill in various professional contexts. Given the dynamic nature of SM platforms and the frequent updates to algorithms, there is a continuous need for leaders to remain interested and engaged with the evolving operational mechanics of each platform. A nuanced understanding of SM dynamics, encompassing both leader and follower actions, offers significant strategic opportunities for those who master these dynamics.

6. Limitations and avenues for future research

In addition to our contributions, our study has limitations that can provide ideas for future research. To begin, we only focused on the top management level of a company. Even though CEOs are considered important actors within an organisation (e.g. Hambrick and Mason 1984; Mackey 2008), other hierarchical levels are also involved in (digital) leadership (e.g. Chief Technology Officer, middle managers) (e.g. Nielsen and Cleal 2011). Further research could therefore investigate how corporate leaders in other hierarchical positions (i.e. vice presidents, directors, managers etc.) or employees with an especially high relevance for a company, within SMEs and start-ups or in specific branches, use the concept of framing via SM and explore potential differences.

Within our research, we further noticed that only around 12% of the CEOs are registered on Twitter, which is a relatively low percentage. This may indicate that CEOs who are active on Twitter systematically differ from the average CEO, which, in turn, constrains the generalizability of our findings. In addition, despite enabling real-time information sharing, Twitter is also characterised by unstructured contributions and a rather short length of tweets (28.71 words per tweet on average), making it hard to analyse content in depth. Additionally, its reputation might motivate a certain behaviour and communication style. Other platforms, such as LinkedIn, do not have such limitations and might also be more frequently used by CEOs. However, data retrieval for research purposes is restricted and partly even forbidden (e.g. web scraping), so it could not be used in our study. Hence, future leadership scholars could additionally analyse framing on other platforms such as LinkedIn (a retrieval might be possible in the future) or picture-based platforms like Instagram.

In addition, leadership as well as SM are contexts in which interaction is vital (Banks et al. 2022). Especially in SM, not only leaders' actions but also followers' reactions are observable, as well as potential dynamics between these two. Therefore, future research should leverage opportunities to examine leadership as a process by simultaneously investigating both leader and follower behaviour, which may precede each other at certain points in time (Güntner et al. 2020).

Also, we explored CEOs context-specific framing on Twitter, compared to non-SM contexts. While we endeavoured to ensure that the data analysed were authentically attributed to the CEOs themselves, particularly in contexts involving public speeches and interviews, it is important to acknowledge a potential limitation: prepared remarks and shareholder letters may not always

be directly penned by the CEOs but could be crafted by speechwriters or public relations experts in consultation with the executive leaders (Matthews, Wang, and Kelemen 2024). This could dilute the ability to accurately capture framing dimensions of the CEOs in our measures.

Despite efforts to mitigate this by focusing on presumably authentic communications (e.g. we removed questions from interviewers to focus solely on CEOs communications), the influence of professional writers in public-facing materials, including SM posts, might negate some observed effects. Nevertheless, this aspect should be considered when interpreting the results of our study.

This paths the way for future research that could further refine our understanding of CEO communications by incorporating methodologies that more directly assess the authenticity of the communication source. This could involve using linguistic analysis tools to differentiate between CEO-written and professionally edited messages. Additionally, expanding the analysis to include a broader range of SM platforms (e.g. LinkedIn, Facebook), as also mentioned by Strauss, Harr, and Pieper (2024), could reveal more nuanced differences in CEOs context-specific framing. This would enhance our comprehension of how leadership styles are conveyed in SM contexts.

Lastly, we have defined framing as a combination of various LIWC dimensions. However, it is not certain that these really cover the entire framing phenomenon. The LIWC dimensions might be only a fragment of actual linguistic framing. Also, we find strong correlations between the concepts of Authenticity and Tone. While they have been introduced as distinct concepts (Boyd and Pennebaker 2015; Tausczik and Pennebaker 2010), in our context, leaders often showed higher Authenticity together with a more positive Tone which makes it hard to methodically distinguish these two concepts. The high correlation might be explained by the fact that spontaneous posts (high Authenticity) on Twitter might be more positive (high Tone) because negative news must be more thought-through to avoid shitstorms (Etter, Ravasi, and Colleoni 2017; Wu et al. 2022), but it is still a limitation that has to be taken into account when discussing our findings. Therefore, we encourage further research to enrich our studies by providing additional methods and investigations to depict these phenomena.

Notes

1. Considering the acquisition of Twitter by Elon Musk in late 2022, the platform has been rebranded as 'X.' However, in our study, we continue to refer to the platform as Twitter, as our data were collected in 2022 before the rebranding. Per the suggestion of an anonymous reviewer, we opted to refer to the platform as 'Twitter/X' in the title of the article to ensure the accessibility of our work for future researchers.
2. Twarc2 is a Python library that facilitates the collection of Twitter data via the Twitter API v2 that can be retrieved from: https://twarc-project.readthedocs.io/en/latest/twarc2_en_us/ (retrieved August 8th, 2024).
3. Socialblade is a website that provides analytics and tracking services for social media platforms and performance related metrics that can be retrieved at: https://socialblade.com/ (retrieved July 28th, 2024).
4. SpaCy is an open-source software library for advanced Natural Language Processing tasks that can be retrieved at: https://spacy.io/ (retrieved July 28th, 2024).
5. Gensim is an open-source software library for unsupervised topic modeling and Natural Language Processing that draws on statistical Machine Learning and can be retrieved at: https://radimrehurek.com/gensim/ (retrieved July 28th, 2024).

Author contributions

CRediT: **Christina Strauss:** Conceptualization; **Michael Dominic Harr:** Data curation; **Torsten Pieper:** Supervision.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendices

Appendix A. Coherence values for topic clusters

Table A1. Topics and coherence values per topic in SM and non-SM contexts.

SM context				Non-SM Context			
No. of topics	CV	No. of topics	CV	No. of topics	CV	No. of topics	CV
1	.421920134	21	.364859788	1	.266680016	21	.435194073
2	.390099641	22	.347832809	2	.402906268	22	.418318765
3	.424293270	23	.370549716	3	.371133074	23	.437737444
4	.413047592	24	.351016609	4	.426086738	24	.434624460
5	.409891850	25	.361232078	5	.409728823	25	.416893592
6	.402713051	26	.344383799	6	.414111243	26	.430758532
7	.415542010	27	.351602428	**7**	**.455180051**	27	.411932184
8	.410259115	28	.348806564	8	.414050353	28	.419652322
9	**.437863157**	29	.352943884	9	.401666972	29	.405034926
10	.422432120	30	.332496829	10	.406614088	30	.423629236
11	.402525084	31	.352039030	11	.406986637	31	.429458685
12	.403169457	32	.341759987	12	.433826167	32	.412895877
13	.397429536	33	.347330686	13	.433519525	33	.402469373
14	.377457459	34	.344452942	14	.420890249	34	.412774964
15	.396444752	35	.354001219	15	.406169048	35	.411466751
16	.380524626	36	.340018141	16	.450873948	36	.429764572
17	.371106590	37	.355504008	17	.412221788	37	.407527160
18	.382700968	38	.340032177	18	.426062352	38	.405330239
19	.377619799	39	.339433304	19	.430683737	39	.413847956
20	.359288078			20	.416532017		

Note: The optimal number of topics (i.e. the maximum C_V) is written bold.

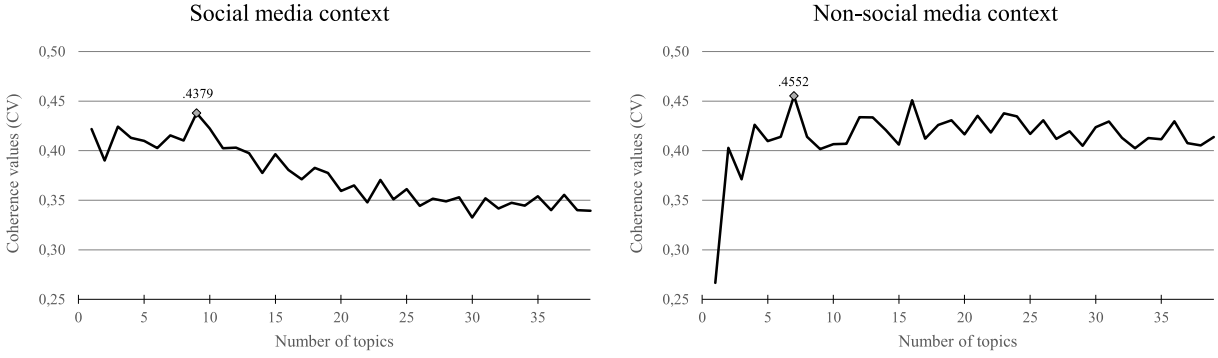


Figure A1. Coherence values for topic clusters in SM and non-SM contexts.

Appendix B. Topic Clusters with most common tokens per topic.

Table B1. Most common tokens per topic in SM and non-SM contexts.

	SM context			Non-SM context		
T	No. of tweets	% of corpus	Top ten tokens per topic	No. of docs	% of corpus	Top ten tokens per topic
T1	13,887	17.31%	.038*'good' + .029*'love' + .025*'read' + .025*'think' + .021*'time' + .020*'know' + .019*'thing' + .017*'okta' + .015*'get' + .015*'want'	71	37.17%	.014*'community' + .012*'health' + .009*'culture' + .009*'impact' + .007*'pandemic' + .007*'ensure' + .007*'goal' + .007*'organization' + .007*'patient' + .007*'crisis'
T2	10,474	13.06%	.156*'great' + .122*'thank' + .063*'team' + .019*'time' + .018*'enjoy' + .017*'customer' + .016*'talk' + .015*'congrat' + .015*'dell' + .015*'today'	40	20.94%	.015*'get' + .013*'lot' + .012*'talk' + .012*'say' + .010*'life' + .010*'like' + .009*'different' + .008*'let' + .008*'happen' + .008*'idea'
T3	9,964	12.42%	.030*'work' + .024*'proud' + .022*'honor' + .022*'world' + .019*'thank' + .019*'year' + .018*'congratulation' + .016*'leader' + .016*'team' + .014*'company'	30	15.71%	.024*'cloud' + .019*'platform' + .016*'digital' + .014*'datum' + .013*'security' + .011*'network' + .008*'organization' + .008*'partner' + .008*'device' + .008*'application'
T4	9,634	12.01%	.024*'customer' + .022*'business' + .019*'new' + .019*'technology' + .017*'help' + .017*'drive' + .016*'innovation' + .014*'industry' + .013*'digital' + .013*'partner'	24	12.57%	.021*'growth' + .020*'quarter' + .017*'strong' + .014*'revenue' + .013*'increase' + .011*'client' + .010*'deliver' + .009*'expect' + .009*'sale' + .009*'term'
T5	8,220	10.25%	.076*'look' + .027*'join' + .027*'welcome' + .025*'week' + .023*'live' + .020*'today' + .019*'event' + .018*'excited' + .017*'congrats' + .016*'new'	16	8.38%	.043*'energy' + .019*'power' + .019*'system' + .015*'emission' + .014*'carbon' + .014*'water' + .013*'vehicle' + .012*'gas' + .011*'investment' + .011*'electric'
T6	8,039	10.02%	.018*'future' + .017*'need' + .016*'change' + .013*'energy' + .013*'step' + .013*'climate' + .012*'world' + .012*'global' + .010*'help' + .010*'action'	5	2.62%	.027*'course' + .020*'try' + .019*'right' + .019*'vaccine' + .018*'lot' + .017*'country' + .015*'price' + .014*'say' + .011*'able' + .010*'percent'
T7	7,497	9.35%	.037*'support' + .035*'happy' + .029*'family' + .027*'help' + .020*'health' + .018*'community' + .016*'thank' + .015*'people' + .015*'year' + .014*'hope'	5	2.62%	.035*'supply' + .026*'chain' + .023*'manufacturing' + .022*'right' + .021*'home' + .018*'china' + .017*'lot' + .017*'term' + .015*'capacity' + .014*'demand'
T8	7,133	8.89%	.064*'new' + .030*'salesforce' + .022*'go' + .019*'cloud' + .018*'video' + .018*'open' + .017*'amazing' + .015*'rt' + .014*'app' + .013*'atlassian'			
T9	5,362	6.68%	.039*'work' + .023*'xyleminc' + .023*'day' + .022*'come' + .020*'water' + .020*'idea' + .020*'long' + .019*'end' + .018*'let' + .018*'happen'			
∑	**80,210**	**100%**		**191**	**100%**	

Note. T = Topic; Percentages have been rounded to two decimal places.